

DTAF AI GUIDELINE

This report is divided into two main complementary parts. The first portion introduces the Digital Twin Analytical Framework (DTAF), including its purpose, structure, and the five integrated methodologies that form its analytical foundation (Sections 1 and 2). The DTAF is designed to help users move from an initial DT idea to a more clearly defined implementation path. The second portion presents the AI-assisted DTAF pipeline developed to simplify the practical use of the DTAF framework (Sections 3 and 4). It provides instructions to guide users through the framework using structured prompts to automatically generate outputs related to capability identification and feasibility, DT maturity recommendations, simulation recommendations, and verification and validation guidance. The DTAF AI pipeline guides software installation, AI environment preparation, workflow execution, user interaction steps, automated processing layers, and interpretation of final outputs.

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1. What is DTAF?

1.1. Overview

While existing frameworks address individual aspects of Digital Twin (DT) implementation, no single approach integrates data readiness, simulation capability, maturity assessment, and trust. The proposed Digital Twin Analytical Framework (DTAF) fills this gap by combining complementary frameworks for each component into a cohesive methodology suitable for practical implementation, enabling small and medium manufacturers (SMMs) to address the challenges of DT adoption systematically. The DTAF aims to offer a strategic roadmap for DT adoption and serve as a means of integrating three digitalization efforts- the Digital Thread, DTs, and advanced simulation. By addressing critical questions surrounding data readiness, scope, fidelity, and frequency, this framework will address the early stages of development, from determining the data elements and capabilities needed to establish the Digital Thread and create a DT, to leveraging the current capabilities to create a DT that can be trusted to support decision-making. To accomplish this, the following five frameworks will be incorporated into DTAF:

- **Digital Twin Consortium (DTC) Capability Periodic Table (CPT):** This framework was selected because it serves as an architecture and technology-neutral tool for mapping requirements and guiding organizations to design from a "use case" perspective rather than focusing on specific features or solutions.
- **Data Element Mapping and Analysis (DEMA):** This framework ensures that all relevant data elements are identified and structured, forming a robust Digital Thread that supports DT implementation.
- **4R Framework:** This framework categorizes DTs into four increasing levels of capability, enabling organizations to plan DT development to match their desired maturity and functional scope, and to align technical effort with business objectives.
- **4S Framework:** This framework provides a clear hierarchy of simulation capabilities, guiding the appropriate scope and utility of simulations within the DT.
- **Digital Twin V (DTV):** This framework applies iterative verification and validation to establish trust and ensure that the DT is ready to support organizations to evaluate whether the DT is trustworthy and ready to guide operational or strategic decisions. Together, these frameworks form a unified methodology that enables systematic, practical, and credible implementation of Digital Twins in manufacturing systems. The DTAF framework components are more thoroughly described below.

The interaction between the frameworks within DTAF is represented in Figure 1 as a two-phase structure: pre-development definition and feasibility (scope, capability intent, requirements framing, and data readiness) feeding staged DT implementation. DTAF is intentionally modular and does not imply that every DT effort requires the full depth of each embedded framework. Rather, the purpose of combining these modules is to provide a structured set of methods that can be selectively applied based on the organization's context, constraints, and targeted DT value.

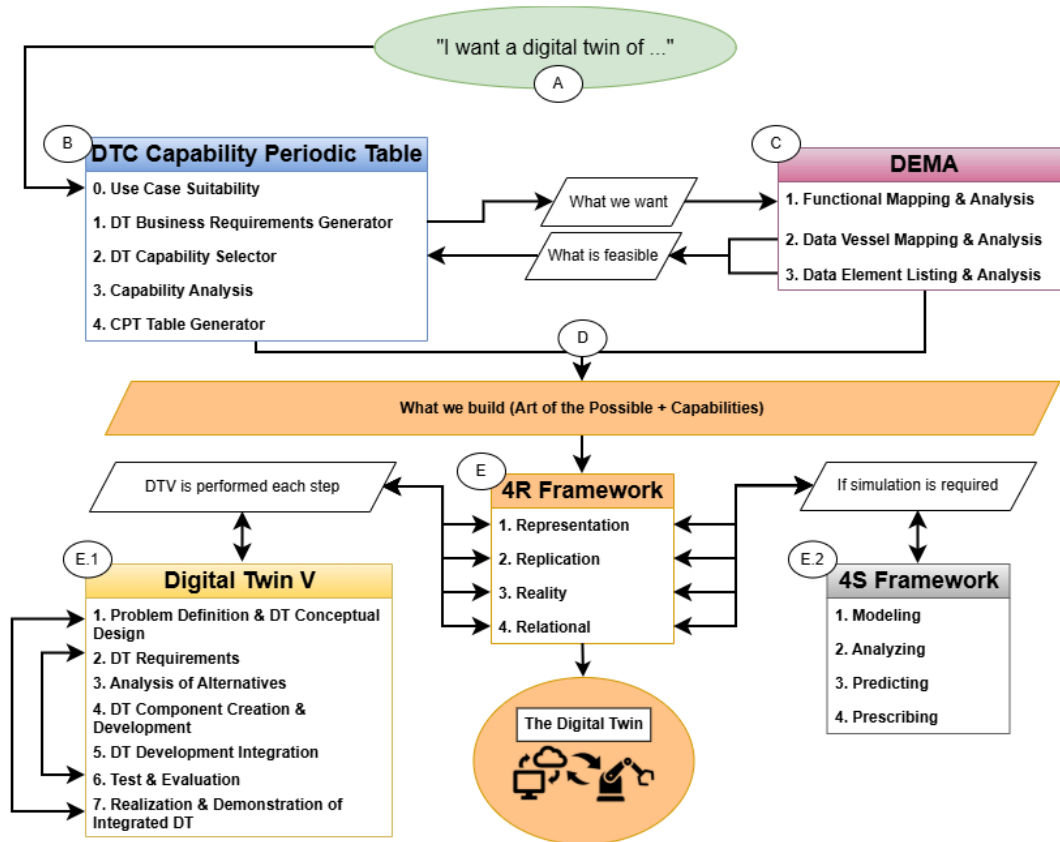


Figure 1: DTAF Structure and Flow.

1.2. Overall Structure and Flow of DTAF

As represented in Figure 1, the DTAF consists of five main steps (A-E), with two sub-steps under Step E (E.1 and E.2). Each step of the DTAF builds upon the previous one to guide practitioners from the initial idea of “I want a DT of...” to an actionable, validated implementation. The steps are described below, and the detailed information on the individual frameworks itself are provided in Section 2.

Step A: Establish the Idea: The process begins with articulating the initial intent behind developing a DT, such as “I want a digital twin of...”. This step helps clarify the motivation, expected value, and relevance of pursuing a DT within a given context. The goal of Step A is to frame the high-level idea before addressing business or technical details. The outcome is a problem intent that serves as the entry point for structured analysis.

Step B: Define Business Context and Requirements: Using the DTC CTP, Step B transforms the initial idea from Step A into a business-anchored problem definition, leveraging toolkits available on the [Digital Twin Consortium GitHub website](#) (user guides, worksheets, and GitHub-hosted resources) so teams can access and apply them directly. The DTC CPT is a technology-agnostic, use-case-oriented framework that organizes the full range of DT

functionalities in a periodic table format, providing a shared vocabulary for designing and benchmarking DT architectures across domains. It enables teams to systematically define what a DT should do, how intelligent it needs to be, and what supporting data or processes are required to achieve its objectives. The AI Agent Capabilities Periodic Table (AIA CPT) extends the CPT framework by integrating AI to automate capability selection, customization, and assessment. It identifies and justifies the specific capabilities required to support the DT use case, then deepens the analysis to emphasize critical operational functions and trustworthiness-related functions that will later anchor verification and validation. The selected DT capabilities that fit the use case are synthesized into a visual periodic table representation that organizes business goals, functional requirements, and capability interdependencies. This produces a picture of what we want to achieve in terms of capabilities and serves as the conceptual starting point for DTAF. The result of Step B is a comprehensive business requirements specification generated by the AI, including stakeholder roles, quantified impacts, DT objectives with KPI targets, data cadence and quality needs, user experience requirements, and a phased implementation plan. This specification provides the foundation for identifying the capabilities a DT must deliver.

Step C: Assess Data Readiness and Feasibility: Once the desired capabilities are identified, Step C evaluates what is feasible using DEMA methodology. DEMA systematically maps the functional, vessel, and data element relationships required to realize the proposed capabilities. By doing so, it exposes hidden data dependencies, verifies data sources and quality, and builds the Digital Thread that links conceptual intent to operational data. The output informs whether the envisioned DT provided from Step B can be realized with the current data maturity or requires further development.

Step D: Integrated Decision to Build: Step D reconciles Step B intent with Step C feasibility evidence by prioritizing capabilities, deferring infeasible items, and documenting required remediation (sensors, interfaces, governance) for later maturity increments. The refined capability architecture and the DEMA evidence are synthesized into a buildable baseline. Step D represents a formal decision point, an evidence-based “go/no-go” gate, to proceed with DT construction. At this stage, the minimal viable set of capabilities is confirmed, data integrations and governance are defined, and success criteria, KPIs, and risks are documented. This controlled decision ensures alignment among business value, technical feasibility, and resource investment, and lays the foundation for the maturity development of the DT.

Step E: DT Development and Maturity Progression: Once feasibility is confirmed, the framework transitions from definition to execution. The 4R Framework is incorporated at this step to provide guidance to the DT development and to add capability levels incrementally. It sequences the development of the DT through the Representation, Replication, Reality, and Relational levels, ensuring each level is a coherent slice of value supported by verified data and validated functions. Within this process, Step E.1 applies DTV as continuous verification and validation steps at each maturity level of the 4R framework, while Step E.2 aligns simulation depth across the 4S framework to the decisions at hand. The result is a controlled path from concept to operations in which scope, fidelity, and risk evolve together.

Step E.1: Verification and Validation: The DTV framework serves as a continuous trust layer integrated into every 4R increment. For each maturity step, it defines the context of use and acceptance criteria, verifies data lineage, synchronization, model structure, and parameterization, and validates performance against KPI targets and error tolerances. Each verification and validation step produces documented test artifacts and evidence logs that trace back to Step D, ensuring consistent trust in DT performance as maturity evolves. Steps B and C establish the capability intent and data feasibility (what is needed and what is possible), whereas Step E.1 does not repeat problem definition or requirements analysis; instead, it operationalizes those outputs by translating them into stage-specific acceptance criteria, verification artifacts, validation activities, and quantitative evidence produced at each 4R maturity increment.

Step E.2: Simulation Alignment: If simulation is considered a capability of the DT implementation, the 4S Framework operates in parallel with the 4R Framework and sets the appropriate simulation depth for each maturity level of the DT, from basic modeling to advanced predictive and prescriptive analytics. The 4S Framework uses inputs from the capability architecture (Step B) and constraints from data readiness (Step C) to provide the levels of capability and complexity required for the simulation component within the DT.

Outcome: The Digital Twin: Through these structured, iterative steps, the DTAF provides a clear pathway from conceptual intent to a trusted, operational DT. Each phase strengthens alignment among business goals, technical feasibility, data integrity, and system trustworthiness, ensuring that the resulting DT is not only functional but also reliable and sustainable.

2. DTAF Frameworks

In this section, each of the frameworks used in the DTAF are explained in detail.

2.1. DTC Capability Periodic Table

Scope and functionality are critical when designing and implementing DTs. A DT must be constructed not as an isolated model or visualization, but as an intelligent, data-driven system that evolves with its physical counterpart and supports real-world decision-making. To achieve this, organizations require structured guidance on defining what a DT should do, how intelligent or autonomous it should be, and which data or processes are necessary to enable that functionality. The Digital Twin Consortium (DTC) provides such guidance through its combined Capabilities Periodic Table (CPT) [1] and AI Agent Capabilities Periodic Table (AIA CPT) [2], two complementary frameworks that together form a comprehensive capability-based design and assessment methodology.

The DTC Capabilities Framework establishes a technology-agnostic, use-case-oriented approach for building DTs and their supporting intelligent agents. Rather than focusing on specific software tools, sensors, or vendors, it emphasizes the capabilities required to achieve a defined business or operational outcome. This shift from tool-centric to capability-centric thinking enables organizations to systematically define the scope and functionality of DT, determining which functions are essential, which are operational, and how these functions evolve as DT matures. Using the periodic table format, the DTC framework maps the full range

of DT capabilities across domains, offering a structured vocabulary for users to collaboratively design and benchmark their digital ecosystems.

At its core, the CPT promotes a capability-driven design approach, in which each use case defines a specific operational or decision-support objective. These use cases are translated into explicit capability requirements using the CPT, which can then be aggregated to determine broader system needs, platform requirements, and technology solutions. This ensures that DT development remains grounded in functional purpose and measurable outcomes, rather than a fragmented, tool-driven implementation.

Building on this foundation, the AIA CPT extends the CPT framework by introducing AI-enabled interactivity to assist in the generation, customization, and assessment of capability models. While the CPT provides a static map of possible DT capabilities, the AIA CPT transforms it into an intelligent interactive toolkit that helps organizations rapidly tailor capability frameworks to their specific use cases. By inputting project details or operational objectives, users can receive AI-generated recommendations for relevant CPT capabilities, prioritized rankings, maturity-level assessments, and implementation roadmaps.

Together, the CPT and AIA CPT form a unified, use-case-oriented framework that bridges the gap between physical systems, digital representations, and cognitive automation. The CPT provides the base map of required capabilities, while the AIA CPT functions as the AI-powered companion that helps generate and operationalize those capabilities within specific organizational contexts. This combination ensures that DT projects are not only well-scoped and technically grounded but also adaptable, scalable, and strategically aligned. By guiding teams from high-level business objectives to detailed capability specifications, the DTC Capability Framework enables systematic, efficient, and trustworthy DT implementation across sectors and maturity levels. Figure 2 illustrates the DTC periodic table.

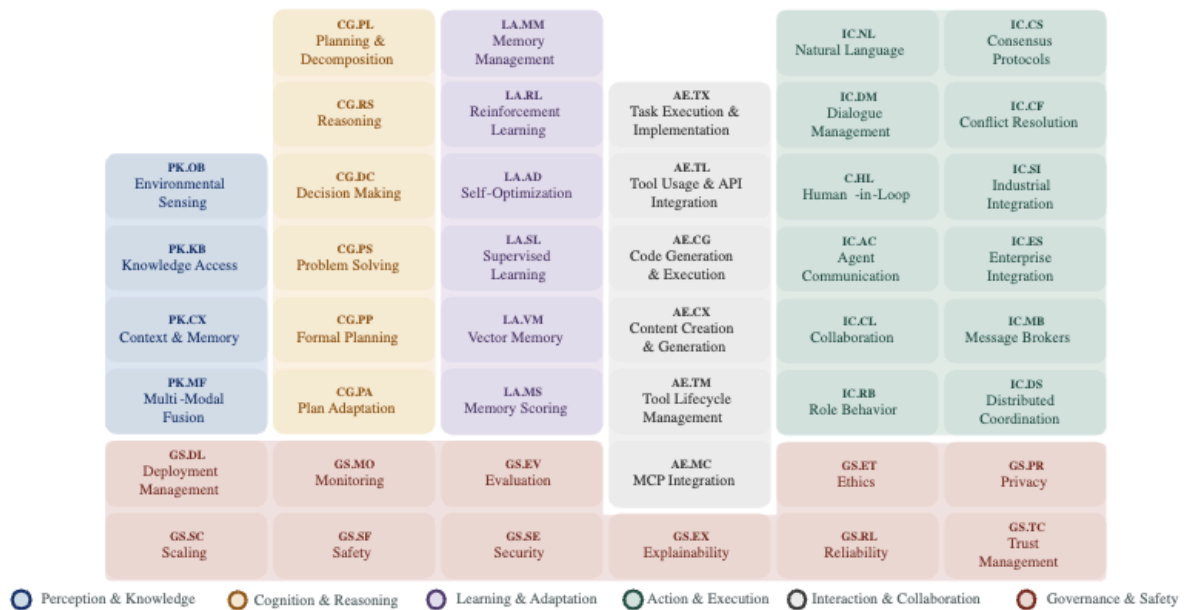


Figure 2: DTC Periodic Table of Digital Twin Capabilities.

2.2. DEMA

Data readiness is foundational for any DT, as unmapped or hidden data flows can compromise its accuracy and utility. Studies have shown that more than 90% of organizational data flows are unknown at the system level, often hidden within siloed knowledge. Therefore, any framework for DT or Digital Thread implementation should account for hidden data and

information flows to accurately reflect real-world systems. To address this, Data Element Mapping and Analysis (DEMA) provides a systematic approach for identifying, capturing, and mapping data flows in complex systems [3,4]. DEMA helps organizations understand where their data truly originates and where it ultimately ends up.

Unlike traditional methods that focus on documents or high-level processes, DEMA dives deeper by examining data at the data element level, the smallest meaningful units, such as part numbers, approval dates, or design dimensions. By doing this, DEMA uncovers hidden or manual data exchanges that often disrupt system efficiency and reliability. It creates transparency by showing how individual data elements connect across departments, software platforms, and lifecycle stages, enabling a more complete and accurate digital thread.

The DEMA process follows three steps. The first step, the Functional-Level Mapping, identifies the major functional areas and sub-activities that make up an organization's workflow. This step establishes the foundation for understanding where data are created, used, and exchanged. Functional-level maps show departments interact and how processes align with organizational goals, helping the analyst trace every piece of data to its functional purpose. The second step, Data Vessel-Level Mapping, dives deeper by documenting the movement of containers, called data vessels, across these functions. Data vessels include spreadsheets, reports, CAD files, emails, or even handwritten notes. It can be anything that stores or transfers data. In this step, DEMA identifies how these vessels are shared, where they are stored, and who has access to them. By capturing these flows, organizations can see how data travels between people and systems and where efficiencies of "dead ends" exist. The third step, Data Element-Level Mapping, is where each data vessel identified in the second step is broken down into its smallest components, individual data elements. These elements might include measurements, serial numbers, inspection results, or timestamps. Mapping these elements allows the analyst to trace exactly how data is created, modified, and reused across systems and departments.

By following these three steps, DEMA transforms hidden, fragmented data into a transparent, traceable network that supports digital engineering initiatives such as DTs. Unlike older mapping tools that focus on functions or documents, DEMA provides a data-centric approach that supports both process improvements and digital transformation. It bridges the gap between systems engineering and lean manufacturing by uncovering inefficiencies, reducing redundant work, and supporting continuous improvement. Figure 3 illustrates the DEMA framework.

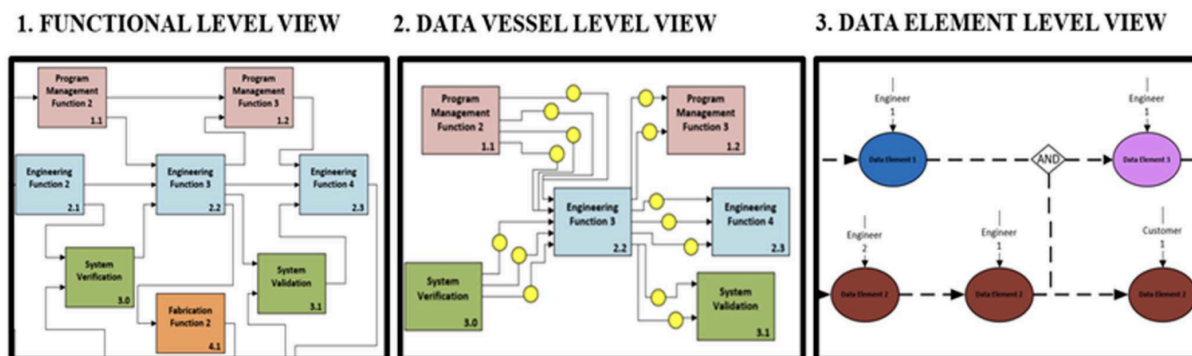


Figure 3: DEMA Overview.

2.3. 4R Framework

DT maturity assessment ensures that the DT is developed to the appropriate level of capability for the intended use case. Since not every use case requires a highly complex DT, organizations must identify the required level of DT capability for their technical requirements, pertinent data, and operational goals. Establishing clear objectives for creating the DT provides clear direction to ensure that the desired performance is achieved. To address this, research was performed to characterize the development of a DT based on its capabilities and maturity as a means of creating general-purpose and modular DTs, called the 4R framework [5,6]. This framework was designed to create a conceptual model that defines and guides the development of the DTs through distinct phases.

The 4R framework categorizes DTs into four phases: Representation (R1), Replication (R2), Reality (R3), and Relational (R4), each representing increasing levels of maturity and capability. The R1 phase involves understanding the physical system's behavior and setting up a data architecture and storage system. The goal is to define the features, behaviors, components, and relationships relevant for the use case, and identify what to include in the model and what not to. In the R2 phase, the system is replicated in a virtual environment using the data architecture established in the representation phase, reproducing the same outputs given the same inputs from the physical system. It is necessary that this phase provides accurate mapping of inputs, as it is fed data from R1 and ultimately becomes the replica used to model the physical system. The R3 phase involves using the DT to explore what-if scenarios, with the aim of implementing the results obtained from virtual runs into the physical system. This creates a virtual modeling tool that can now run independently of the physical system with the ability to make predictions, provide alternative scenarios, and optimize. The R4 phase involves incorporating decision-making technologies such as AI or ML and enabling bi-directional data exchange. This allows the model to translate updates, corrections, and decisions back into the physical system, making the twin self-aware.

Each level of the 4R Framework builds upon the previous one, progressively increasing the model's analytical depth and decision-making capability. As capability advances from R1 to R4, the DT moves from the ground level of data gathering and component identification to full autonomy. This hierarchical structure demonstrates the progression from analyzing and applying data to build a digital replica, training that replica to optimize the physical system, and ultimately enabling the DT to make independent updates and corrections to the physical system, which form the foundation requirements of DT implementation. By clearly defining these stages, the 4R Framework enables researchers and practitioners to assess their desired DT capabilities, identify existing gaps, and determine the necessary level of support needed for their engineering objectives. Ultimately, the 4R framework establishes a systematic approach for transitioning from data architecture development to the creation of an intelligent, data-driven DT capable of continuous feedback, optimization, and real-time decision support. Figure 4 illustrates this ordered progression and highlights how increasing complexity correlates with the enhanced ability of the DT.

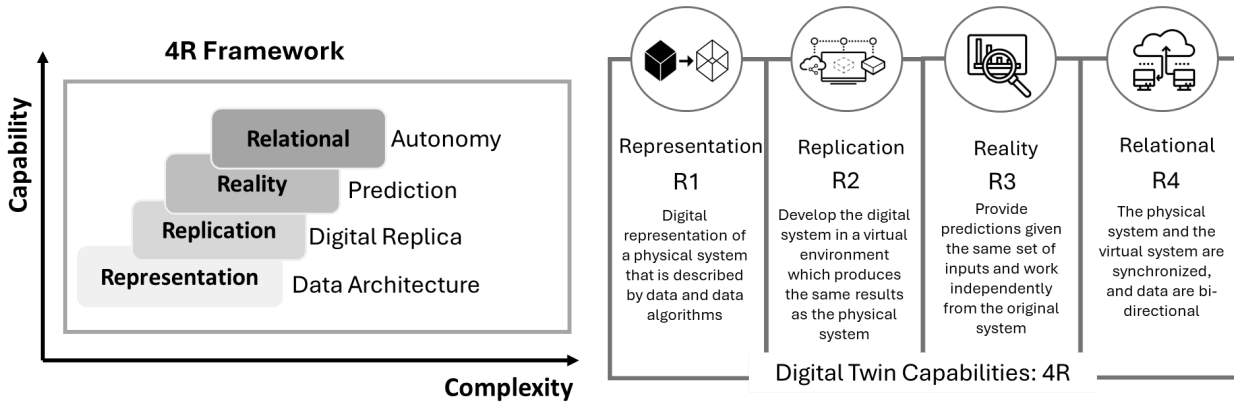


Figure 4: 4R Framework.

2.4. 4S Framework

Simulation capability is another critical aspect. Correctly assessing simulation levels prevents mislabeling simulations as DTs. Traditional simulation and DT models share the same capability to replicate physical systems in virtual environments, but they are not the same. Simulation is at the core of DTs, and their combined use is very common, creating confusion: many organizations mistakenly label simulation models as DTs and vice versa. While simulations have their own valuable applications, it's inaccurate to classify them as DTs. Unlike simulations, DTs feature continuous bilateral communication, sensor-based monitoring of physical system changes, and real-time decision-making support. Research has found that a significant portion of literature claiming to combine simulation and DT fails to achieve true DT capabilities, instead presenting simulations as DTs. Even among correctly built DTs, many only utilize a fraction of their potential. When constructing a DT, it is important to identify the simulation's capabilities and ensure they fit the specific scope for each individual problem. Current literature has addressed this by categorizing simulation into four levels of capability and utility into one framework - The 4S Framework [7,8].

The 4S framework comprises 4 levels: Modeling (S1), Analyzing (S2), Predicting (S3), and Prescribing (S4). The S1 level involves creating an abstracted representation of a real-world system through conceptual modeling and computer modeling. Proper data collection, parameter selection, and understanding are crucial in this stage. In the S2 level, the data gathered during Modeling is incorporated into the simulation model. Input analysis ensures that the data used approximates real-world behavior. At this level, analysis is conducted to infer the system's behavior. The S3 level is where the simulation can predict system behavior under various conditions. It allows for the investigation of what-if scenarios, performance measurement, and comparison. The S4 level involves using the simulation model for decision support and optimization. Decision-makers can identify optimal courses of action, and simulation optimization or machine learning methods can be applied to enhance decision-making and system performance.

Each level of the 4S Framework builds on the previous one, progressively increasing the model's analytical depth and decision-making value. As capability advances from S1 to S4, the simulation evolves from a descriptive to a prescriptive system. This hierarchical structure not only illustrates the growth of analytical complexity but also aligns simulation maturity with the foundational requirements for DT implementation. By clearly defining these stages, the 4S Framework enables researchers and practitioners to assess their current simulation readiness, identify capability gaps, and determine the appropriate level of sophistication needed to support specific organizational or engineering objectives. Ultimately, the 4S framework provides a

structured pathway for transitioning from isolated simulation models to intelligent, data-driven systems capable of continuous feedback, optimization, and real-time decision support. Figure 5 illustrates the simulation levels of capabilities and complexity.

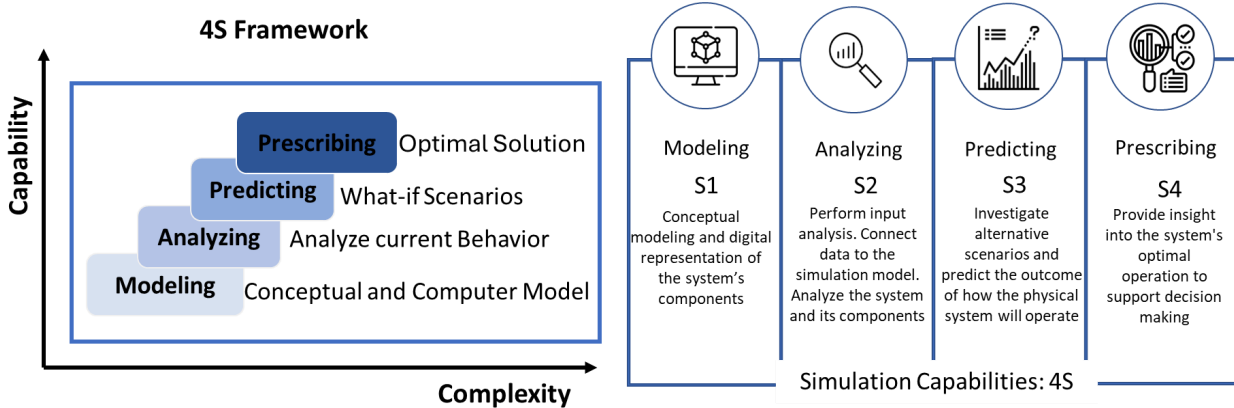


Figure 5: 4S Framework.

2.5. Digital Twin V

Finally, trust and credibility are essential for a DT to support real-world decision-making. It is fundamental to trust the data, accuracy, and updating procedure of the models that support decision-making, especially in manufacturing systems where cost and safety are critical. Therefore, Verification and Validation (V&V) are crucial pillars of creating trust and credibility in DTs. In this context, establishing trust includes demonstrating how the V&V evidence can be applied to support the model's use in a defined context of use. From an investment perspective, building trust in the DT implementation minimizes the risk associated with technology adoption and acceptance. One existing framework that addresses this gap is the Digital Twin V (DTV) [9, 10]. The DTV is a methodology inspired by the Vee model introduced by Forsberg and Mooz (1991), and it describes in a structured approach how to create trusted DTs by applying V&V throughout the DT development life cycle. The framework consists of iterative, recursive steps that create a methodology for considering the elements needed to develop trusted, integrated DTs.

The DTV breaks down into 3 stages: decomposition, development, and integration and testing. In stage 1, decomposition, the problem is defined and broken down into DT conceptual design and technical requirements. The conceptual design outlines the model structure and describes how the physical system will be mirrored digitally. The technical specifications define the DT's purpose, context of use, and data requirements. Stage 2, development, begins with the analysis of alternatives, which involves creating and implementing solutions that fulfill the previously determined capability and complexity requirements. Stage 3 is integration and testing. At this stage, the DT will be deployed to the operational environment and continuously monitored. Continuous feedback loops are put in place, running a series of tests to ensure real-time feedback and recalibration. Verification ensures each iteration of the DT fulfills the design and technical requirements. There are two types of verification: model verification ensures the DT's components meet performance requirements, while code verification includes static and dynamic condition testing. Simultaneously, validation compares each iteration's performance and accuracy against the real system's performance, using comparison experiments that compare DT outputs to real system data, or thorough uncertainty quantification that measures aleatory and epistemic uncertainties. Upon the conclusion of verification and validation, the DT development is complete. The DTV requires documentation of V&V evidence to demonstrate trustworthiness, reproducibility, and transparency. The three stages of the DTV

establish a replicable process that embeds V&V in every phase of DT development and provides a baseline for ongoing updates and long-term model optimization. Figure 6 shows the structure and stages of the DTV, with the V&V steps emphasized.

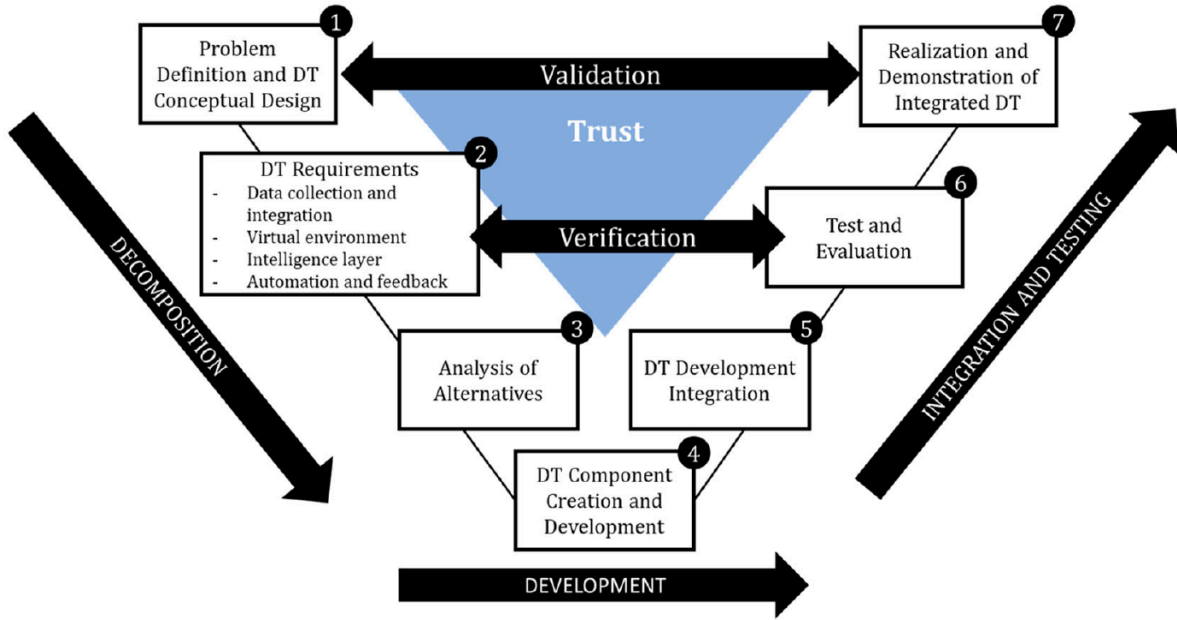


Figure 6: Digital Twin V.

3. DTAF AI Pipeline

While the DTAF provides a comprehensive and structured methodology for guiding DT adoption, its multi-framework nature can pose a practical challenge for SMMs. Many SMMs operate with limited technical personnel, constrained budgets, and minimal time available to interpret multiple frameworks, determine how they relate to one another, and translate them into actionable implementation steps. To address this challenge, an Artificial Intelligence (AI)-assisted pipeline was developed to support the practical deployment of the DTAF. The purpose of the pipeline is to simplify navigation through the framework, reduce the time users spend actively, and improve consistency in how the methodology is applied.

The pipeline follows the logical sequence of the DTAF, beginning with capability identification through the Digital Twin Consortium (DTC) Periodic Capability Table to establish organizational context, operational needs, and target use cases. From there, the AI assists users in progressing through subsequent stages, including data element identification, maturity classification, simulation capability assessment, and trust considerations related to verification and validation. At each stage, the system prompts the user for relevant inputs, interprets responses, and structures outputs in a format aligned with the selected framework. This approach reduces the burden of manually studying each framework independently and minimizes the likelihood of overlooking key considerations. In summary, the AI pipeline serves as an enabling layer for the DTAF, converting a comprehensive yet potentially complex methodology into a guided, efficient, and secure implementation tool.

3.1. Preparing the AI Pipeline Environment

To support privacy and secure deployment, the pipeline utilizes **Ollama** with the **Qwen2.5:7B** large language model running locally on the user's machine. Local execution is particularly important for manufacturing organizations that may need to discuss sensitive operational data, proprietary workflows, production bottlenecks, or strategic priorities. Because the model runs locally, organizational information does not need to be transmitted to external cloud-based AI services, reducing concerns regarding data leakage, confidentiality, or unauthorized reuse of proprietary information.

To achieve reliable performance when running the AI pipeline locally, the following minimum and recommended hardware specifications are suggested in Table 1 below.

Table 1: Minimum and Recommended Computer Requirements.

Component	Minimum	Recommended
RAM	16 GB	32GB+
CPU	4 modern cores	8+ cores
GPU	6 to 8 GB VRAM	12 to 16 GB VRAM
Storage	20 GB SSD	30+ GB NVMe SSD
OS	Windows 10/11	Windows 11

3.2. Required Software Installation

STEP 1: Install Python: Python is required to run the pipeline.

- Visit the official Python website: <https://www.python.org/downloads/>
- Download the latest version.
- Run the installer and complete the installation.

STEP 2: Install Visual Studio Code (Recommended): Although any Python interpreter may be used, Visual Studio Code (VS Code) is recommended.

- Download VS Code: <https://code.visualstudio.com/>
- Install the software.
- Open VS Code.
- Open VS Code, go to the Extensions icon on the left sidebar, search for "Python," and click Install.

STEP 3: Install Ollama: Ollama allows the AI model to run locally on your computer to protect sensitive company data because information is not sent to external cloud servers.

- Visit: <https://ollama.com/>
- Download Ollama.
- Install the software.
- Open Ollama once after installation.

STEP 4: Install the AI Model: The DTAF pipeline uses Qwen2.5:7B. This model was selected because it provides strong reasoning performance while remaining manageable on standard hardware.

- Open Command Prompt on your system by pressing **Windows key + R**
- Type: **cmd** and press **Enter**
- Copy and paste the following command into the terminal and press Enter: **ollama pull qwen2.5:7b**. This downloads the model for the first time.
- Wait until installation completes.

STEP 5: AI DTAF GitHub: The files and materials needed to run the AI DTAF are within the GitHub repository.

- Visit the GitHub repository that contains the DTAF AI pipeline files: <https://github.com/lkw0012-sudo/AI-DTAF/tree/main>
- Click the provided project package named: **DTAF.zip** and download the zip folder. See Figure 7 below.

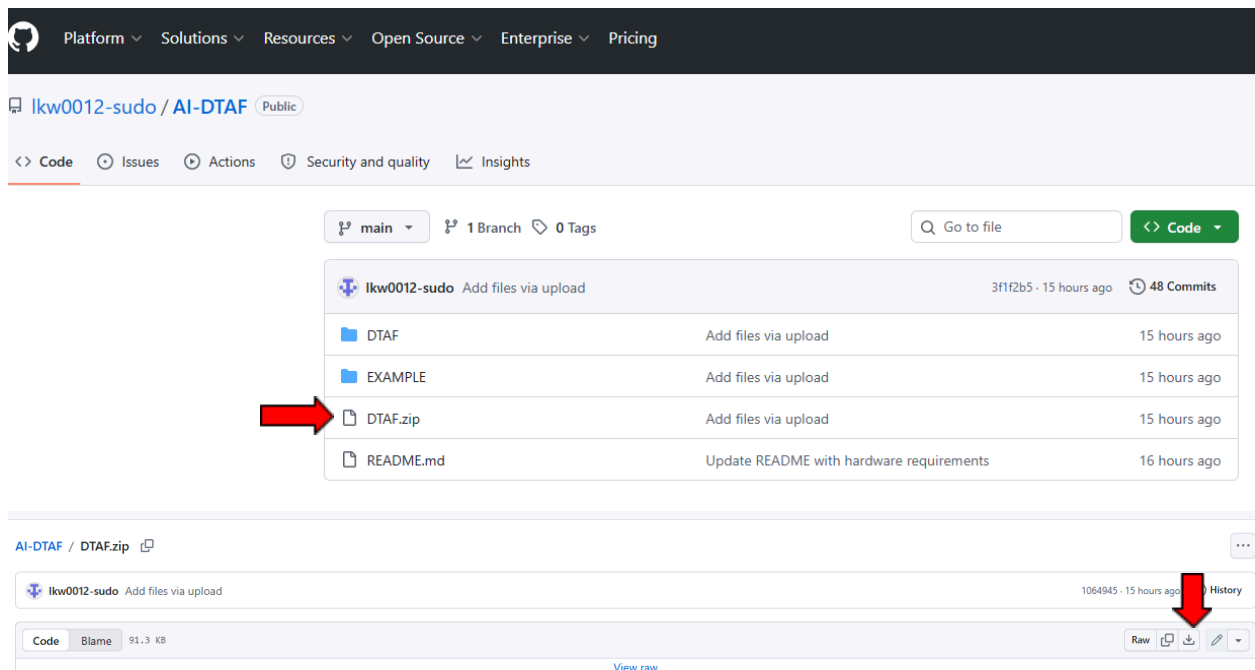


Figure 7: Downloading the DTAF.zip folder from GitHub.

- Go to the Downloads folder, right-click the ZIP file, and select: Extract All
- Open the extracted folder. This is where all the required files are located.

- *Important:* Do NOT remove files from the extracted folder. All files must remain together. If the main Python file is moved elsewhere, the program may fail because it depends on internal folders and knowledge-base files.
- Inside the extracted folder (Figure 8), you will see:
 - JSON Files: These serve as the AI knowledge base and contain logic for the DTAF stages, capability mappings, decision rules and framework guidance
 - Markdown (.MD) Files: These contain reference material from the Digital Twin Consortium, including: capability descriptions, prompt guidance, and capability Periodic Table definitions
 - Python File (.py): This is the main executable file that the user will run, called **DTAF.py**

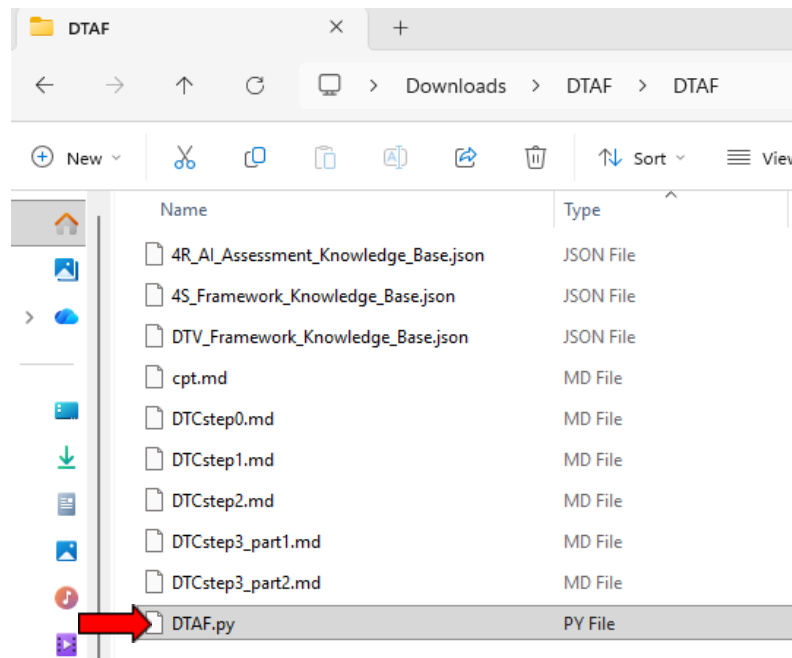


Figure 8: Extracted DTAF.zip folder.

STEP 6: Running the Program:

- In the extracted folder, Double-click on **DTAF.py** to run the Python file within VS Code.
- The Python file will look like the following (Figure X) when opened with VS Code.
- Click the Run button identified with the red arrow in Figure 9.

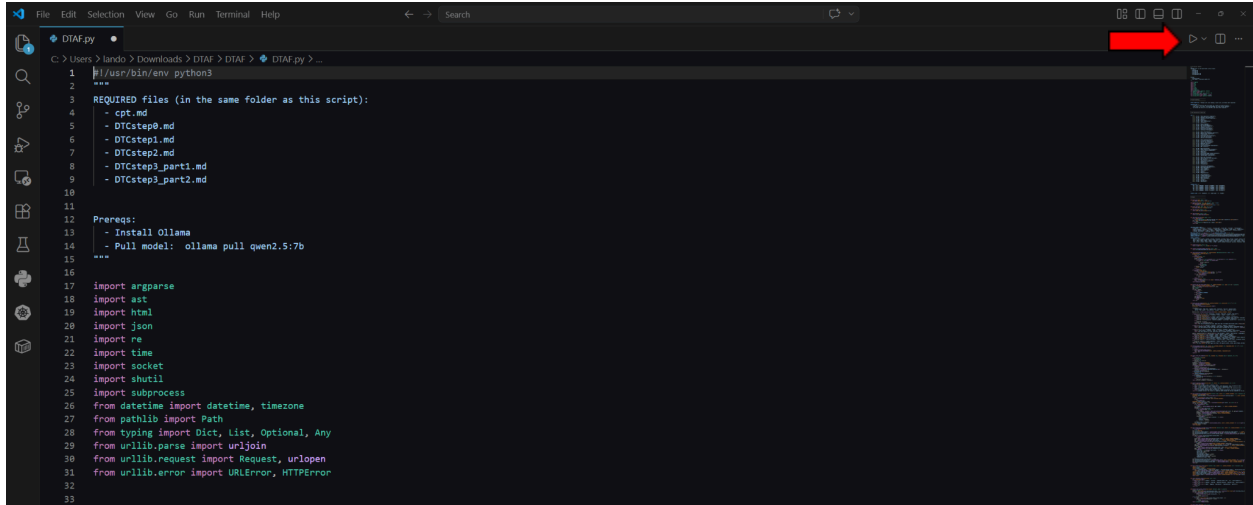


Figure 9: Python file opened in VS Code.

4. DTAF Workflow Using the AI Pipeline

This section explains the operational workflow of the DTAF AI pipeline and the interaction between the user and the AI system. As illustrated in Figure 10, the architecture is structured into five layers that align with the logic of the DTAF methodology. Through these layers, the AI progressively converts an initial user-defined problem statement into structured recommendations related to DT capabilities, maturity level, simulation requirements, and trustworthiness verification and validation needs.

The workflow is divided into two major phases: the User-Driven Phase and the Automated Analytical Phase. The User-Driven Phase captures the contextual knowledge and decision inputs required from the user. It consists of the DTC Capability Periodic Table Layer and the Data Feasibility Layer. In this phase, the user defines the sector and problem statement, while the AI interprets these inputs to identify relevant business requirements, essential DT capabilities, and capability gaps. Additional user responses are then used to assess whether the required capabilities are currently feasible based on available data, system functions, and organizational readiness.

Once the foundational inputs are established, the process transitions into the Automated Analytical Phase, which no longer requires direct user interaction. This phase includes the 4R + Build Instruction Layer, 4S Layer, and DTV Layer. Here, the AI analyzes the outputs from the previous phase against the predefined databases to determine the recommended DT maturity level, identify readiness gaps, assess whether simulation functionality is required, and define the necessary verification and validation activities to ensure trustworthiness of the proposed DT solution. The pink boxes represent the knowledge databases associated with the frameworks utilized within the DTAF methodology, serving as the structured knowledge sources that guide the AI reasoning process at each stage. The green boxes represent the outputs generated by the AI after analyzing user inputs. These outputs are consolidated into a single comprehensive HTML file containing the recommendations, assessments, maturity evaluations, readiness gaps, and implementation guidance needed for the user to develop the DT. A detailed explanation of each layer and its associated workflow is provided in the following sections.

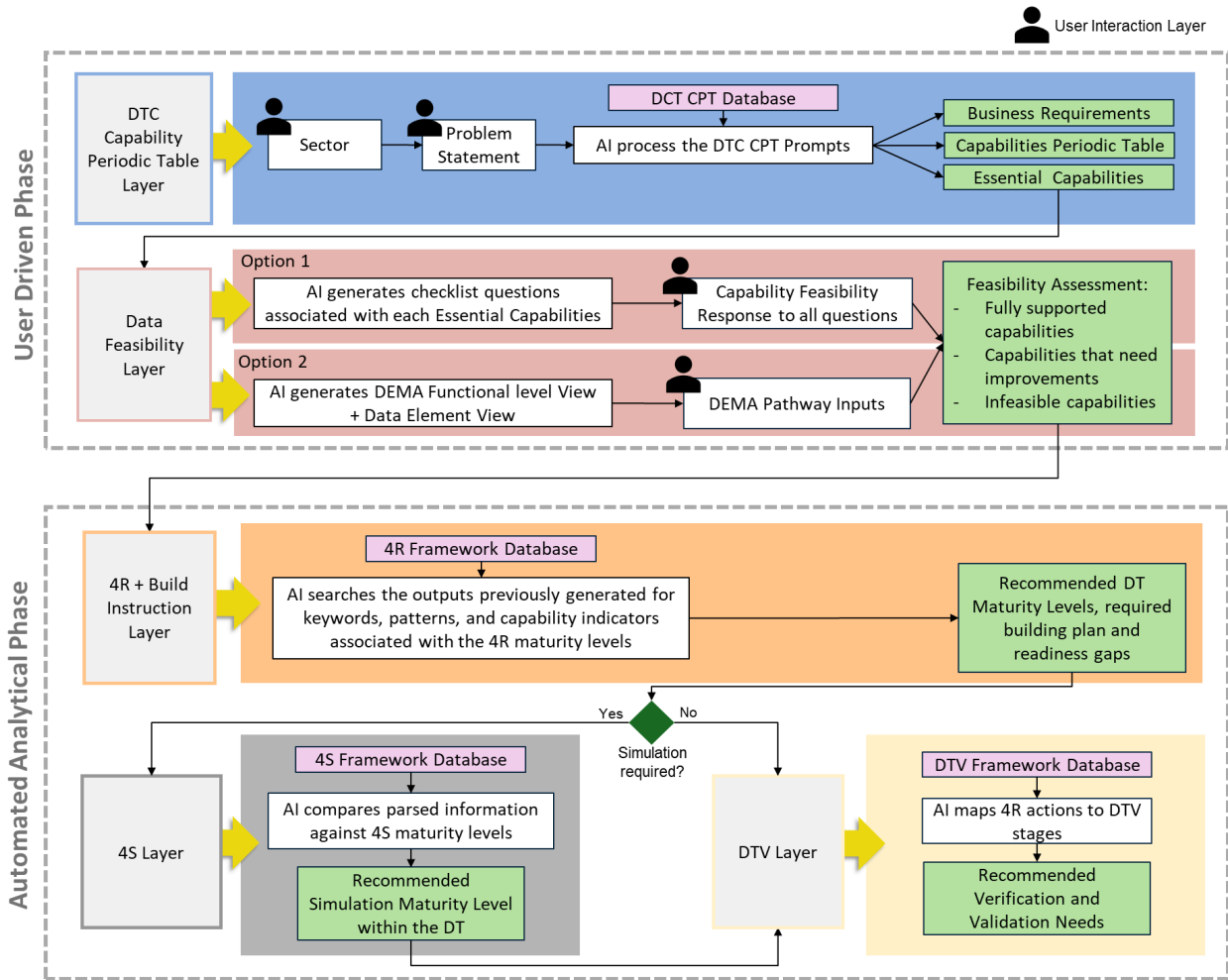


Figure 10: DTAF AI Pipeline Workflow.

The AI pipeline is intentionally designed so that only a limited portion of the workflow requires direct user interaction. Users primarily provide four forms of input:

- Industry sector
- Problem statement / DT objective
- Capability feasibility responses
- Optional DEMA pathway inputs

Once these inputs are complete, the AI automatically executes all remaining stages. User interaction within these layers is explained in the following subsections.

4.1. Digital Twin Consortium Capability Table Layer

The pipeline begins with the Digital Twin Consortium Capability Periodic Table Layer, which serves as the foundation for identifying the capabilities required to support the proposed Digital Twin use case. At the start of this layer, the AI prompts the user with the question: “What

sector should be analyzed?” This is referred to as **Step 0** of the DTC CPT layer. The user should enter the industry sector of interest and then press **Enter**. Examples of sectors include manufacturing, aerospace, logistics, and healthcare.

Once the sector is entered, the AI proceeds to the next prompt and requests a **problem statement** describing the intended Digital Twin use case. This statement should clearly explain the system of interest, the operational challenges it addresses, and the Digital Twin's desired objectives. A well-defined problem statement is critical because it guides the AI in selecting relevant capabilities and generating meaningful downstream recommendations. An example of a complete problem statement is: *“We need a digital twin of our CNC machining systems to provide accurate, data-driven insight into Quality, Cost, and Time before production begins. The digital twin must determine whether we have the capability and capacity to run a given job, optimize scheduling based on available equipment, and automate key decision-making processes that currently rely on manual judgment. It should continuously monitor the degradation of cutting tools, spindles, and coolant, predict remaining useful life, and assess how component wear will impact part quality and throughput. By integrating real-time machine data with predictive simulation, the digital twin will enable proactive maintenance, reduce cycle time, extend asset life, and ensure that every job is run on the right machine, at the right time, with predictable and efficient performance.”*

After completing the problem statement, press **Shift + Enter** to move to a new line, type **END** in capital letters, and then press **Enter**. This signals to the AI that the input is complete and allows the pipeline to begin processing the DTC CPT prompts. This may take 1-10 minutes, depending on hardware. The purpose of this phase is to interpret the organizational context and generate initial outputs related to business requirements, relevant digital capabilities, and operational needs. These outputs establish the baseline understanding required for later stages of the pipeline.

4.2. Data Feasibility Layer

The second stage of the pipeline consists of two alternative pathways within the Data Feasibility Layer: **Option 1** and **Option 2**. These two paths provide different methods for assessing whether an organization is prepared to support the implementation of the proposed DT capabilities.

Option 1 focuses on a data feasibility checklist, which assesses whether the capabilities identified in the DTC CPT layer are realistically achievable given the organization's current resources, infrastructure, and data environment. This option is intended to provide a rapid readiness assessment by identifying which critical capabilities are already supported and which may require additional development.

Option 2 follows a guided Data Element Mapping and Analysis (DEMA) process. This pathway provides a more detailed, systematic approach to identifying, capturing, and mapping data flows across the system. Specifically, it examines the system from three complementary perspectives: the Functional-Level View, the Data Vessel-Level View, and the Data Element-Level View. For users interested in completing Option 2, please refer to the separate document titled “DTC-DEMA AI Guideline” for detailed instructions.

This report focuses on describing Option 1. In this pathway, the AI first extracts all essential capabilities identified during the previous DTC CPT stage. It then generates a series of user-facing checklist questions associated with each capability. The total number of questions presented will depend on how many essential capabilities were identified from the earlier prompts.

The user is then asked to respond to each question by indicating which supporting elements currently exist within their organization or physical system. These responses allow the AI to evaluate the readiness and feasibility of implementing each required capability. An example of one such checklist question is shown in Figure 11.

```
Capability: DS.AI (E) – Data Acquisition & Ingestion
Notes: Ensure all data sources are reliable and high-quality to support real-time decision-making.
A) Real-time data acquisition from IoT sensors [E] [HARD]
B) Integration with ERP and CMMS systems [E]
C) High-quality data sources [E] [HARD]
D) Latency less than 5 seconds for critical data [E]
E) Scalability to handle over 50 CNC machines [E]
F) Data quality checks in place [H]
G) Real-time monitoring of machine health and tool usage [H]
H) Edge computing capabilities for real-time processing [H]
Answer with letters you HAVE (e.g., A C). Or: all / none / skip
A: A

Capability: IR.ET (E) – Enterprise System Integration
Notes: Critical for seamless integration and real-time data flow between digital twin and enterprise systems.
A) Integration with ERP system [E] [HARD]
B) Integration with CMMS system [E]
C) Real-time data streaming from IoT devices [H] [HARD]
D) Data consistency across systems [H]
E) Support for multiple manufacturing plants [H]
Answer with letters you HAVE (e.g., A C). Or: all / none / skip
A: B C
```

Figure 11: Example of a Question from the Capability Feasibility Checklist.

To answer each capability checklist question, the user should type the letter(s) corresponding to the items that currently exist within their physical system or organization. Each letter (A, B, C, etc.) represents a specific capability element, resource, or condition listed in the prompt. The user should select only the options that accurately reflect their current state. For convenience, the system accepts the following responses:

- A B E (example)– If only certain items apply, the user should enter the relevant letters separated by spaces.
- ALL – if all listed items are currently available
- NONE – if none of the listed items are available
- SKIP – if the user is unsure or wishes to move to the next question without answering

After entering the response, press Enter to submit and proceed to the next question. The AI will continue presenting checklist questions until all essential capabilities identified for the DT use case have been assessed. Users should repeat this process until the full capability evaluation is complete. Based on the responses, the AI ranks each capability according to readiness and feasibility. This produces a feasibility assessment, which serves as a screening

mechanism to identify fully supported capabilities, capabilities requiring improvement, and currently infeasible capabilities. This layer ensures that later recommendations are grounded in realistic implementation conditions rather than ideal assumptions.

Upon completion of all user-driven inputs, the AI-assisted DTAF pipeline transitions to an automated analytical stage, where three integrated framework layers are executed to generate a complete DT development recommendation. At this point, no additional user interaction is required. During this stage, the system processes all previously collected information, including sector context, problem statement, essential capabilities, and feasibility responses, and executes three integrated framework layers: the **4R Digital Twin Layer**, **4S Simulation Layer**, and **Digital Twin V (DTV) Layer**. These layers operate automatically without requiring additional user input. Their purpose is to convert the user's responses into a structured implementation roadmap covering DT maturity, simulation requirements, and trustworthiness validation. These layers determine:

- Recommended DT maturity levels, required actions, and readiness gaps (**4R Layer**)
- Recommended simulation maturity level within the DT (**4S Layer**)
- Recommended verification and validation needs (**DTV Layer**)

Given the recommended hardware requirements, a user would expect the model to complete the three layers within 5 minutes. The combined results are compiled into a final output package containing recommendations, readiness gaps, and implementation guidance.

4.3. 4R Digital Twin Layer

The third stage of the pipeline is the 4R Layer, which determines the most appropriate DT maturity level target for the use case and translates that target into a build path. Using a dedicated 4R framework knowledge database, the AI parses all validated information from the essential capabilities and searches for keywords, patterns, and capability indicators associated with the four maturity levels: Representation (R1), Replication (R2), Reality (R3), and Relational (R4). After the comparison, the AI identifies the highest level of maturity reasonably supported by the available evidence and generates a recommended build path, including readiness gaps and actions needed to progress further.

4.4. 4S Simulation Layer

Once the target 4R level is identified, the AI evaluates what type of simulation capability is needed to support that DT. Using the 4S database provided within the DTAF AI, the system compares the parsed information against the four simulation levels: Modeling (S1), Analyzing (S2), Predicting (S3), and Prescribing (S4). The AI classifies the minimum simulation level required to effectively support the recommended 4R maturity level. It then generates guidance for achieving the selected simulation stage.

4.5. DTV Layer

The final stage of the pipeline is the DTV layer, which focuses on trust, testing, and evidence collection for the proposed DTV. Using the DTV database and the outputs from the previous layers, the AI maps the generated action items to DTV stages and then produces verification and validation checks, required evidence, and measurable acceptance criteria.

4.6. Outputs of the DTAF AI Pipeline Process

Although the three analytical layers run independently, their results are merged into a single final output package in .html format. This output includes:

- Overview of the use case
- Recommended 4R maturity level
- Recommended 4S simulation level
- A table of the recommended DT capabilities and the feasibility of each
- Readiness gaps
- Build roadmap
- Verification & validation plan

The primary output generated is the file “digital_twin_build_guide.html”. This serves as the main deliverable for users to review, share with stakeholders, save for future reference, or print as a planning document. It consolidates all recommendations and action items produced throughout the AI pipeline into a single, user-friendly report. Specifically, it contains the actions required to build a DT to the recommended maturity level, achieve the necessary simulation capability level, and establish an appropriate verification and validation strategy to ensure the resulting DT can be trusted for decision-making. Figure 12 shows an example of the front page of the final output, which includes a high-level breakdown of the recommended DT capabilities, 4R and 4S levels, and overall guidance for DTV. Several additional supporting files are also generated and linked within the final HTML report for users who require deeper technical analysis. These supplementary documents include intermediate outputs generated by the model at each layer of the AI pipeline.

4R Build Target

Recommended target: R2 Replication

Current evidence baseline: R1 Representation

Near-term next: R3 Reality

The planned digital twin for CNC machining systems is focused on providing real-time data-driven insights and automated decision-making processes to optimize scheduling, maintenance, and quality control. The feasible capabilities identified support the R2 level of the 4R Framework, which involves replicating the physical system in a virtual environment.

4S Simulation Guidance

Recommended 4S level: S2 Analyzing

Simulation required: Yes

The current evidence supports modeling and analyzing the CNC machining systems to provide real-time data-driven insights and automated decision-making processes, which aligns with S2 (Analyzing) level of simulation capability.

DTV Trust Guidance

DTV should act as a verification, validation, and trust-evidence layer that supports the recommended R2 build target while first closing the lower-level trust-evidence gaps reflected by the current evidence baseline of R1. For this CNC Machining Systems use case, DTV should connect each lower-level and current-level 4R action item to concrete checks, evidence artifacts, and measurable acceptance criteria using data and records from CNC machine sensors, Maintenance logs, Quality control systems, Production scheduling software, Historical job data where applicable.

How to Use This Guide

- Start with the overview cards to confirm the feasible capabilities, 4R build target, 4S simulation level, and DTV trust focus.
- Use the Capability Periodic Table as the visual anchor for what capabilities were selected and prioritized.
- Use the 4R section as the main build path, the 4S section as simulation guidance, and the DTV section as verification and validation guidance.
- Open the supporting Step 1 to Step 3 sections only when you need deeper traceability or more detailed rationale.

Capability View (Step 4 CPT Table)

This section preserves the original Step 4 visual as closely as possible. When available, the script generates an image preview from the standalone Step 4 HTML so the periodic table keeps its intended appearance.

Digital Twin Capabilities Periodic Table V1.2							
Complete Framework of 60+ Core Capabilities							
ESSENTIAL DS.AI Data Acquisition & Ingestion	HIGH VALUE DS.SG Synthetic Data Generation	ESSENTIAL IR.ET Enterprise System Integration	ESSENTIAL IC.SR Search	HIGH VALUE IC.PR Prediction	ESSENTIAL UX.BV Basic Visualization	HIGH VALUE UX.DB Dashboards	
ESSENTIAL DS.ST Data Streaming	FUTURE DS.ON Ontology Management	ESSENTIAL IR.EG Eng. System Integration	— IC.CC Command & Control	ESSENTIAL IC.AI Artificial Intelligence	HIGH VALUE UX.AV Advanced Visualization	— UX.CI Continuous Intelligence	
ESSENTIAL DS.TR Data Transformation	FUTURE DS.RP Digital Twin (DT) Model Repository	ESSENTIAL IR.IO OT/IoT System Integration	ESSENTIAL IC.OS Orchestration	FUTURE IC.PS Prescriptive Recommendations	ESSENTIAL UX.RM Real-time Monitoring	— UX.BI Business Intelligence	
HIGH VALUE DS.CX Data Contextualization	— DS.IR DT Instance Repository	HIGH VALUE IR.DT Digital Twin Integration	— IC.AL Alerts & Notifications	FUTURE IC.FL Federated Learning	— IC.BR Business Rules	— UX.ER Entity Relationship Visualization	— UX.BP BPM & Workflow
ESSENTIAL DS.BP Batch Processing	— DS.DS Domain Specific Data Management	FUTURE IR.CL Collab Platform Integration	— IC.RP Reporting	FUTURE IC.SM Simulation	— IC.DL Distributed Ledger & Smart Contracts	— UX.XR Extended Reality (AR/VR/MR)	FUTURE UX.GE Gaming Engine Visualization
ESSENTIAL DS.RT Real-time Processing	— DS.SA Data Storage & Archive Services	— IR.AS API Services	— IC.AA Data Analysis & Analytics	FUTURE IC.MA Mathematical Analytics	— IC.CS Composition	FUTURE UX.GM Gamification	FUTURE UX.3R 3D Rendering
ESSENTIAL DS.AS Asynchronous Integration	FUTURE DS.SR Simulation Model Repository	ESSENTIAL MG.DM Device Management	HIGH VALUE MG.EL Event Logging	FUTURE TW.EX Data Encryption	ESSENTIAL TW.SC Security	FUTURE TW.SF Safety	FUTURE TW.RP Responsibility
HIGH VALUE DS.AG Data Aggregation	— DS.AR AI Model Repository	ESSENTIAL MG.SM System Monitoring	FUTURE MG.DG Data Governance	FUTURE TW.DS Device Security	HIGH VALUE TW.PR Privacy	FUTURE TW.RL Reliability	FUTURE TW.RS Resilience

Figure 12: Example of what the front page of the primary output looks like.

5. Conclusion

The DTAF addresses a significant gap in the DT implementation by integrating five complementary methodologies (the DTC Capability Periodic Table, DEMA, the 4R framework, the 4S framework, and the DTV) into a single structured decision framework. Rather than treating data readiness, capability analysis, DT maturity, simulation fidelity, and trust verification as isolated concerns, DTAF unifies them into a singular analytical toolbox. The DTAF AI pipeline demonstrates how AI can operationalize a comprehensive DT planning methodology into a practical, accessible decision-support tool. Applying each framework within DTAF manually can require significant time, expertise, and coordination. The AI pipeline addresses this challenge by transforming the framework into a guided, automated workflow that reduces the user burden while preserving the methodology's analytical rigor. Through guided prompts, automated reasoning, and structured outputs, the system significantly reduces the time and expertise traditionally required to evaluate and plan DT initiatives.

Together, the DTAF methodology and AI pipeline enable SMMs to develop trusted and capable DTs without requiring expertise in each underlying framework within DTAF. MxD will deploy the DTAF methodology and AI pipeline as a resource available to its members, supporting broader adoption across the manufacturing community. Future enhancements may include expanded domain customization, improved user interfaces, stronger integration with live enterprise data systems, and continuous learning capabilities to further strengthen decision support and accelerate the adoption of DTs.

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